

Introduction to Unsupervised Learning - Notes

1. Unsupervised Learning

Learns patterns from unlabeled data. The algorithm discovers hidden structures without target labels.

2. High Dimensionality

Data may contain many features (dimensions). Each feature is one dimension.

3. Vector Encoding

Categorical values are converted into vectors. Example:

Mumbai → [0,1,0,0,...]

Pune → [1,0,0,0,...]

This is similar to one-hot encoding.

4. What is a Vector?

A vector is an ordered list of values representing one data point.

Example: [80, 33, 9, 22000] = IQ, Age, Experience, Salary.

5. Features / Dimensions

Each column (Age, Salary, etc.) is called a feature or dimension.

6. Curse of Dimensionality

As dimensions increase:

- Data becomes sparse.
- Distance becomes less meaningful.
- Computation becomes expensive.
- Overfitting risk increases.

7. Sparsity

Most values are zero or empty. Sparse data can lead to overfitting if not handled properly.

8. Coefficient of Variation (CV)

$CV = \text{Standard Deviation} / \text{Mean}$

Low CV → High consistency.

High CV → Low consistency.

9. Concentration of Measure

In very high dimensions, distances between points become similar, reducing the usefulness of distance metrics.

10. Cosine Similarity

Measures similarity by comparing the angle between vectors. Values range from -1 to 1; closer to 1 means more similar.

11. PCA (Principal Component Analysis)

A dimensionality reduction technique that keeps the most important information while reducing the number of features.

Exam Tips

- Vector = numerical representation of data.
- Curse of dimensionality affects distance-based algorithms.
- PCA reduces dimensions while preserving maximum variance.
- Cosine similarity is widely used for text embeddings and recommendation systems.